Answer Key & Teacher Notes

Session 1 — Building a Histogram · Session 2 — Describing Histogram Shape

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Building a Histogram

TALK PROMPTS

- Why must the intervals be the same width and never overlap?
- How can you check your frequency table before you draw any bars?

PARTNER TASK

Pushups. One partner tallies the pushup data into 0–9, 10–19, 20–29 while the other draws the touching bars. Swap roles to check: the bar heights must add up to the number of students.

WATCH FOR

- Using overlapping intervals such as 5–10 and 10–15, so a value is counted twice.
- Leaving gaps between bars as in a bar graph.
- Reading the tallest bar as one value that repeats instead of a whole interval.

SESSION 2 · TEACHER NOTES

Describing Histogram Shape

TALK PROMPTS

- What does the peak of a histogram tell you about a typical value in the group?
- How can two data sets with the same number of values have such different shapes?

PARTNER TASK

Two Histograms. Partner A sketches bars 2, 6, 9, 6, 2 and Partner B sketches 10, 6, 4, 3, 2 (25 values each). Each names the other's shape and peak, then they explain together why the same count can give different shapes.

WATCH FOR

- Naming skew by where the tall bars are instead of by the tail (calling 9, 5, 2, 1, 1 skewed left).
- Thinking a bar shows one exact value that repeats.
- Calling any histogram with a middle peak symmetric without matching the bars on each side.

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.SP.B.4	1	1	1	2	1	2	2	3	2	1	1
S1 · Form B	6.SP.B.4	1	1	1	2	1	2	2	3	2	1	1
S2 · Form A	6.SP.B.4	2	1	2	1	2	2	3	3	2	1	1
S2 · Form B	6.SP.B.4	2	1	2	1	2	2	3	3	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

histogram: A graph of **touching** bars; each bar counts the values in one interval.

interval: A group of numbers, like 10–19. All intervals have the same **width**.

frequency: **How many** values land in one interval.

- Intervals must be the same **width** and must not **overlap**.
- The bar heights add up to the **total** number of values.

Example: 0–9 holds 4 and 9, so its frequency is **2**. 10–19 holds 15, 11 and 18: frequency **3**. 20–29 holds 22 and 27: frequency **2**. Answer: Check: 2 + 3 + 2 = **7** values

YOU TRY

1. a. 1 b. 3 c. 4

2. 10 − 6 = 4

EXIT TICKET

1. a. 2 b. 3 c. 2

2. 15 − 10 = 5

3. **A** (0–4, 5–9, 10–14)

PRACTICE FORM A

1. 0–9: 2 10–19: 4 20–29: 3; touching bars of height 2, 4, 3. Most rides: 10–19

2. a. 1 b. 3 c. 4 d. 8

3. a. 20 b. 50–59

4. 9 − (2 + 4) = 3

5. a. 1 b. 2 c. 2 d. 2

6. The intervals overlap: 10, 15 and 20 each fit in two intervals. Model: 5–9, 10–14, 15–19, 20–24.

7. 50–59, 60–69, 70–79, 80–89, 90–99

8. Table: 30–39: 5 40–49: 4 50–59: 3. a. 30–39 b. The bars show at a glance where most times fall; a list of 12 numbers does not.

9. Histogram bars must touch because the intervals leave no gaps on the number line. Correct: three touching bars of height 1, 2, 1 (4 is in 0–9; 12 and 18 are in 10–19; 26 is in 20–29).

10. **B** (frequency of the interval)

11. **C** (17)

Self-check word **TALLIES**

PRACTICE FORM B

1. 0–9: 3 10–19: 4 20–29: 2; touching bars of height 3, 4, 2. Fewest rides: 20–29

2. a. 1 b. 3 c. 5 d. 9

3. a. 22 b. 40–49

4. 12 − (3 + 5) = 4

5. a. 1 b. 2 c. 2 d. 3

6. The intervals overlap: 5, 10 and 15 each fit in two intervals. Model: 0–4, 5–9, 10–14, 15–19.

7. 40–49, 50–59, 60–69, 70–79, 80–89

8. Table: 30–39: 5 40–49: 5 50–59: 2. a. 50–59 b. The bars show at a glance where most times fall; a list of 12 numbers does not.

9. Histogram bars must touch because the intervals leave no gaps on the number line. Correct: three touching bars of height 1, 2, 1 (4 is in 0–9; 12 and 18 are in 10–19; 26 is in 20–29).

10. **C** (0–9, 10–19, 20–29)

11. **B** (leave no gaps between them)

Self-check word **COUNTED**

GUIDED NOTES

peak: The interval with the **tallest** bar.

symmetric: The left and right sides **match**.

skewed right: The long **tail** stretches to the right.

skewed left: The long tail stretches to the **left**.

- The **peak** shows where a typical value falls.
- Name a skewed shape for its **tail**, not for where the tall bars are.

Example: The tallest bar, 9, is over **30–39**, so that is the peak. The bars shrink toward the **left**, so the tail is on the left. Answer: The shape is **skewed left**.

YOU TRY

1. a. symmetric b. skewed right

2. 70–79; 15 values

EXIT TICKET

1. a. symmetric b. skewed left

2. 20–29; 16 values

3. **B** (6 values are from 40 to 49)

PRACTICE FORM A

1. a. 2 b. 2 c. 4 d. 1 e. 20–24 f. skewed left (the tail stretches left)

2. a. 1–2 lb b. 14 c. skewed right (the tail stretches right toward 7–8 lb)

3. a. 30–39 b. skewed right (the tail stretches right toward 50–59)

4. a. 70–79 b. 12

5. a. symmetric b. skewed right c. skewed left

6. Yes. The bars are 1, 2, 9, 2, 1: each bar left of the peak matches the bar the same distance to its right.

7. 2 peaks (0–9 and 30–39 are both tallest). The bars on the two sides of the middle do not match (6, 2, 1, 6, 2), so it is not symmetric.

8. A histogram. Almost no value repeats, so a dot plot would be a long row of single dots. Intervals group the values so the shape shows.

9. The tallest bar stands for a whole interval, not one value. It means 9 values fall somewhere in that interval; the histogram does not show the exact values.

10. **C** (skewed right)

11. **B** (tallest bar)

Self-check word **SKEWED**

PRACTICE FORM B

1. a. 2 b. 2 c. 2 d. 4 e. 25–29 f. skewed left (the tail stretches left)

2. a. 7–8 lb b. 15 c. skewed left (the tail stretches left toward 1–2 lb)

3. a. 20–29 b. skewed right (the tail stretches right toward 50–59)

4. a. 60–69 b. 17

5. a. symmetric b. skewed left c. skewed right

6. Yes. The bars are 2, 3, 8, 3, 2: each bar left of the peak matches the bar the same distance to its right.

7. 2 peaks (0–9 and 30–39 are both tallest). The bars on the two sides of the middle do not match (5, 2, 1, 5, 3), so it is not symmetric.

8. A histogram. Almost no value repeats, so a dot plot would be a long row of single dots. Intervals group the values so the shape shows.

9. The tallest bar stands for a whole interval, not one value. It means 7 values fall somewhere in that interval; the histogram does not show the exact values.

10. **C** (symmetric)

11. **A** (skewed left)

Self-check word **SHAPES**

SHEET 1 · BUILDING A HISTOGRAM

1. 0–9: 3 (5, 8, 9); 10–19: 6 (11, 12, 14, 16, 17, 19); 20–29: 3 (21, 23, 25). Total 12.
2. Bars of height 3, 6, 3. a. 10–19 b. 9 games
3. a. 3 (22, 27, 29) b. 4 (31, 34, 36, 38) c. 3 (41, 45, 48). Total 10.
4. 24 − (5 + 9 + 3) = 24 − 17 = 7 students
5. The intervals are not equal widths (10, 5, and 15 values). Better: 0–9, 10–19, 20–29.

SHEET 2 · DESCRIBING HISTOGRAM SHAPE

1. a. 21 students b. 80–89 c. Skewed left d. 14 students
2. a. Symmetric b. Skewed right c. Skewed left d. Symmetric e. Skewed right f. Skewed left
3. a. 3 + 8 = 11 runners b. 6 + 2 = 8 runners
4. Model: 1, 2, 3, 6, 8 (sum 20). Any 5 heights with sum 20 whose tallest bars are on the right and that shrink toward the left.
5. No. The histogram only shows that 8 students scored from 80 to 89. It does not show the exact scores.
6. **B** (10, 7, 4, 2, 1)
7. **C** (12)

October 2026 conceptual additions — teacher key

Session 1 · Building a Histogram

Frequencies are 2, 2, 3, 1. Ten belongs only in 10–19 and twenty only in 20–29. Eight observations are counted exactly once.

Extension: Use $0 \le t < 10$, $10 \le t < 20$, $20 \le t < 30$, and $30 \le t < 40$. Then 9.5 belongs in the first interval.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Describing Histogram Shape

The tail extends left, toward the lower-frequency intervals. The highest bar is at the right, but skew is named for the tail, so this display is left-skewed.

Extension: One example is 1, 3, 3, 1. Mirrored frequencies give symmetry and total 8.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.